Classmate

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PHYSICS

Q1) 4) +0.5

Q2) (2) Outer surface of the cornea

Q3) (1) Real, inverted, diminished

Q4) (1) Both A and R are True and R is the correct explanation of A

Q5) (2) Both assertion and reason are true but reason is not the correct explanation of assertion

Q6) The two common defects of vision are Myopia and hyper Ametropia. Myopia can be corrected by using a ~~convex l~~ concave lens and hypermetropia can be corrected by using convex lens.

Q7)(a) Water is optically denser than air ~~&~~. The light from the part of pencil, which is immersed in water travels from denser to rarer medium i.e. water to air, the light bends away from the normal and the pencil appears to be bent and shorter

(b) $m = -2$

$v = +20$

$f = ?$

$m = \frac{v}{u}$

$-2 = \frac{20}{u}$

$u = \frac{\cancel{20}^{10}}{\cancel{-2}} \Rightarrow u = -10cm$

Using lens formula

$\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$

$\frac{1}{20} - \left(\frac{1}{-10}\right) = \frac{1}{F} \Rightarrow \frac{1}{20} + \frac{1}{10} = \frac{1}{f}$

$\Rightarrow \frac{1+2}{20} = \frac{1}{f} \Rightarrow \frac{3}{20} = \frac{1}{f}$

$\Rightarrow f = \frac{20}{3} \Rightarrow f = +6.67 cm$

08) $f = +15\,cm$
$u = -10\,cm$

Using mirror formula.

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{15} = \frac{1}{v} + \frac{1}{(-10)} \Rightarrow \frac{1}{15} = \frac{1}{v} - \frac{1}{10}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{15} + \frac{1}{10} \Rightarrow \frac{1}{v} = \frac{2+3}{30}$$

$$\Rightarrow \frac{1}{v} = \frac{\cancel{5}\,1}{\cancel{30}\,6} = \frac{1}{6} \Rightarrow v = 6\,cm$$

Nature of the image is Virtual and erect. The image is formed 6 cm behind the mirror

09) $u = -10\,cm$

$$m = -\frac{v}{u} \qquad \cancel{v = +3\,cm \times}$$

$$\cancel{3 = \frac{-v}{}} \quad v = +3 \times (-10) = -30\,cm.$$

Using lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{f} = \frac{-1}{30} - \left(\frac{1}{-10}\right)$$

$$\frac{1}{f} = -\frac{1}{30} + \frac{1}{10} \Rightarrow \frac{1}{f} = \frac{-1+3}{30}$$

$$\Rightarrow \frac{1}{f} = \frac{2}{30} \Rightarrow f = 15\text{ cm.}$$

Q10) The person is suffering from Myopia and needs a concave lens to correct the defect

$\therefore\ f = -x = -80\,cm$

$= -0.8\,m.$

$Power = \frac{1}{f}$

$\Rightarrow \frac{1}{(-0.8)} \Rightarrow \frac{1}{-8} \times 10.$

$= \frac{-10}{8} = -1.25\,D$

Q11) (a) The lens forms a real image so it is a convex lens.

(b) Negative as the image is real and inverted

(c) $f = 20\,cm$
$u = -21\,cm$

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\frac{1}{20} = \frac{1}{v} - \left(\frac{-1}{21}\right)$

$\Rightarrow \frac{1}{20} = \frac{1}{v} + \frac{1}{21}$

$\frac{1}{v} = \frac{1}{20} - \frac{1}{21}.$

$$\frac{1}{v} = \frac{21-20}{420} \Rightarrow \frac{1}{v} = \frac{1}{420}$$

$v = 420\,cm$

Q12) (a) Laws of refraction are :-

(i) The incident ray, refracted ray & normal to the interface of two media at the point of incidence, all lie in the same plane

(ii) the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant

absolute refractive index =

$$\frac{\text{Speed of light in vaccum}}{\text{Speed of light in medium}} = \frac{c}{v}$$

in terms of $\angle i$ and $\angle r$

$$n = \frac{\sin i}{\sin r}$$

1

(b) Refractive index of water = $\frac{4}{3}$

Refractive index of glass = $\frac{3}{2}$ 1

Speed of light in glass = $2 \times 10^8\,ms^{-1}$ 1

(i) Speed of light in air

$$= \frac{2 \times 10^8 \times 3}{2}$$

$$= \boxed{3 \times 10^8 \ ms^{-1}}$$

(ii) Speed of light in water

$$= \frac{c}{n_w}$$

$$= \frac{3 \times 10^8 \times 3}{4}$$

$$= \boxed{2.25 \times 10^8 \ ms^{-1}}$$